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To cite this version:

Jean-Guillaume Dumas, Clément Pernet, Alexandre Sedoglavic. A non-commutative algorithm for multiplying a 3x4 matrix by a 4x7 matrix using 63 non-complex multiplications. 2025. ⟨hal-05121550v2⟩

HAL Id: hal-05121550
https://hal.science/hal-05121550v2

Preprint submitted on 15 Jul 2025

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A non-commutative algorithm for multiplying a 3×4 matrix by a 4×7 matrix using 63 non-complex multiplications

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July 15, 2025

Abstract

The number of scalar multiplications required to multiply a 3×4 matrix by a 4×7 matrix was reduced from 66 in [12] to 63 in [10] by an algorithm over complex numbers.

We propose an algorithm in 63 multiplications with only rational coefficients, hence removing the requirement that the base field for the arithmetic operations contains a square root of -1, while it still needs to contain an inverse of 2. It was derived from the latter one, under the action of an isotropy which happen to project the algorithm on the field of rational numbers. We also produce a straight line program of this algorithm.

1 Introduction

V. Strassen introduced in [13] the first sub-cubic time algorithm for matrix product, with 7 recursive multiplications and, thus, a complexity of $O(n^{2.81})$.

Recently, an algorithm to multiply a 3×4 matrix by a 4×7 requiring only 63 scalar complex-valued multiplications was found using a pipeline of large language models orchestrated by an evolutionary coding agent [10].

We present in this note a *equivalent variant* of this algorithm that only use rational coefficients and no longer rely on complex values: in fact, we consider the *tensor decomposition representation* encoding the original algorithm and an *isotropy* that sends it to another decomposition naturally with the same *tensor rank* but that does not involve complex coefficients anymore.

The next section is devoted to present the framework allowing this journey. Section 3 gives a *trilinear representation* of the tensor decomposition exhibited in this note. Furthermore, it presents the isotropy used to obtain it and the stabilizer of the considered orbits. Section 4 gives some straight line program that implements the presented theoretical algorithm.

2 Tensor decomposition encoding matrix product

In this note, we will use the representation of matrix multiplication algorithm given by a tensor decomposition of the corresponding bilinear map. For a more detailed discussion of this framework, we refer the reader to [9]. Nevertheless, we expose it following the illustrative example of Strassen's matrix multiplication algorithm.

Illustrating example Consider three matrices:

$$\mathsf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathsf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad \mathsf{C} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}. \tag{1}$$

The product $\mathsf{C} = \mathsf{A} \cdot \mathsf{B}$ of 2×2 matrices can be obtained using Strassen's algorithm using the following computations:

$$\begin{array}{ll} \rho_1 \leftarrow a_{11}(b_{12} - b_{22}), & \rho_4 \leftarrow (a_{12} - a_{22})(b_{21} + b_{22}), \\ \rho_2 \leftarrow (a_{11} + a_{12})b_{22}, & \rho_5 \leftarrow (a_{11} + a_{22})(b_{11} + b_{22}), \\ \rho_3 \leftarrow (a_{21} + a_{22})b_{11}, & \rho_7 \leftarrow (a_{21} - a_{11})(b_{11} + b_{12}), \\ \rho_6 \leftarrow a_{22}(b_{21} - b_{11}), & \left[\begin{smallmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{smallmatrix}\right] = \left[\begin{smallmatrix} \rho_5+\rho_4-\rho_2+\rho_6 & \rho_2+\rho_1 \\ \rho_6+\rho_3 & \rho_5+\rho_7+\rho_1-\rho_3 \end{smallmatrix}\right]. \end{array} \tag{2}$$

This straight-line program (a.k.a. SLP) encodes the following bilinear map over a field $\mathbb{R}$ with m, k, n — that are all equal to 2 in Strassen's algorithm:

$$\begin{array}{llll} \beta_{\text{MM}}: & \mathbb{R}^{m \times k} \times \mathbb{R}^{k \times n} & \to & \mathbb{R}^{m \times n}, \\ & (\mathsf{A}, \mathsf{B}) & \mapsto & \mathsf{A} \cdot \mathsf{B}. \end{array} \tag{3}$$

Indices m, k and n are used in this section for clarity, helping to easily distinguish the different spaces involved in the following sections such as $\mathbb{R}^{m \times k}$ the space of $m \times k$ matrices with coefficients in $\mathbb{R}$.

2.1 Matrix multiplication encoded by tensors decomposition

Following [9], we begin by expressing the bilinear map defining matrix product into the formalism of tensor decomposition, that provides a clear way to represent algorithms associated to such product and its associated symmetries.

Strassen's algorithm (2) is encoded as the decomposition of the matrix multiplication tensor, expressed as a sum of seven rank-one tensors $\mathcal{S}$ defined by the following relations:

$$\begin{aligned} \mathcal{S} = \textstyle\sum_{i=1}^{7} \mathsf{M}_i \otimes \mathsf{N}_i \otimes \mathsf{O}_i = & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ + \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + & \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ + \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} + & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \\ + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + & \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \end{aligned} \tag{4}$$

in $\left(\mathbb{R}^{m \times k}\right)^{\star} \otimes \left(\mathbb{R}^{k \times n}\right)^{\star} \otimes \mathbb{R}^{m \times n}$ (with $m = k = n = 2$ in Equation (4)).

Definitions 2.1 *Given $\mathcal{T}$ a tensor decomposition $\sum_{i=0}^{r} \mathsf{T}_{i1} \otimes \mathsf{T}_{i2} \otimes \mathsf{T}_{i3}$ where each summand is a* rank-one tensor*:*

- *the integer r is the* tensor rank *of tensor decomposition $\mathcal{T}$;*
- *the ordered list $\left[(\operatorname{rank} \mathsf{T}_{ij})_{j=1\ldots3}\right]_{i=1\ldots r}$ is called the* type *of tensor decomposition (*rank M *being the classical rank of the matrix M). The type could be summarized as the multivariate polynomial $\sum_{i=0}^{r} X^{\operatorname{rank} \mathsf{T}_{i1}} Y^{\operatorname{rank} \mathsf{T}_{i2}} Z^{\operatorname{rank} \mathsf{T}_{i3}}$.*

In the above Strassen's tensor decomposition, the *tensor rank* is 7 and its type is $X^2Y^2Z^2 + 6XYZ$.

Furthermore, we are going to use in Section 2.2 and Section 3 the following relationship between the classical trace of linear operator and tensor representations introduced so far. The Frobenius inner product $\langle\ \rangle$ of a matrix O with a matrix product $\mathsf{M} \cdot \mathsf{N}$ defines a trilinear form $\operatorname{Trace}(\mathsf{O}^{\intercal} \cdot \mathsf{M} \cdot \mathsf{N})$ as follows:

$$\begin{array}{llll} \mathcal{S}|_3: & \mathbb{R}^{m \times k} \times \mathbb{R}^{k \times n} \times (\mathbb{R}^{m \times n})^{*} & \to & \mathbb{R}, \\ & (\mathsf{M}, \mathsf{N}, \mathsf{O}^{\intercal}) & \mapsto & \langle \mathsf{O}, \mathsf{M} \cdot \mathsf{N} \rangle. \end{array} \tag{5}$$

The complete contraction of this tensor $\mathcal{S}$ with an *input* tensor $\mathsf{A}\otimes\mathsf{B}\otimes\mathsf{C}$ is its *trilinear representation*:

$$
\begin{aligned}
&(a_{11}+a_{22})\otimes(b_{11}+b_{22})\otimes(c_{11}+c_{22}) && (6)\\
&+(a_{12}-a_{22})\otimes(b_{21}+b_{22})\otimes c_{11} && (7)\\
&+(-a_{11}+a_{21})\otimes(b_{11}+b_{12})\otimes c_{22} && (8)\\
&+(a_{11}+a_{12})\otimes b_{22}\otimes(-c_{11}+c_{12}) && (9)\\
&+a_{11}\otimes(b_{12}-b_{22})\otimes(c_{12}+c_{22}) && (10)\\
&+a_{22}\otimes(-b_{11}+b_{21})\otimes(c_{11}+c_{21}) && (11)\\
&+(a_{21}+a_{22})\otimes b_{11}\otimes(c_{21}-c_{22})\,. && (12)
\end{aligned}
$$

We present our result in Section 3 using this kind of representation.

2.2 Symmetries of matrix product tensor decomposition

Remark that the matrix product is associated to $\mathrm{Trace}(\mathsf{A}\cdot\mathsf{B}\cdot\mathsf{C})$ by Equation (5) and that, given three invertible matrices $\mathsf{U},\mathsf{V},\mathsf{W}$ of suitable sizes and the classical trace properties, this trace is equal to:

$$
\begin{aligned}
&\mathrm{Trace}\big((\mathsf{A}\cdot\mathsf{B}\cdot\mathsf{C})^{\intercal}\big)=\mathrm{Trace}(\mathsf{C}\cdot\mathsf{A}\cdot\mathsf{B})=\mathrm{Trace}(\mathsf{B}\cdot\mathsf{C}\cdot\mathsf{A})\\
&\text{and to }\ \mathrm{Trace}\big(\mathsf{U}^{-1}\cdot\mathsf{A}\cdot\mathsf{V}\cdot\mathsf{V}^{-1}\cdot\mathsf{B}\cdot\mathsf{W}\cdot\mathsf{W}^{-1}\cdot\mathsf{C}\cdot\mathsf{U}\big).
\end{aligned}
\tag{13}
$$

These relations illustrate the next theorem and induce the isotropy action on matrix product tensor decomposition presented below:

Theorem 2.1 ([6, § 2.8]) *The* isotropy group *of the $m\times k$ by $k\times n$ matrix multiplication tensor is the semidirect product*

$$\left(\mathrm{PSL}^{\pm}\left(\mathbb{R}^{m}\right)\times\mathrm{PSL}^{\pm}\left(\mathbb{R}^{k}\right)\times\mathrm{PSL}^{\pm}\left(\mathbb{R}^{n}\right)\right)\rtimes\mathfrak{S}_3, \tag{14}$$

where PSL *stands for the group of matrices of determinant* ±1 *and* $\mathfrak{S}_3$ *for the symmetric group on* 3 *elements.*

As we do not harness the $\mathfrak{S}_3$ part of this isotropy group, we do not give further details on its action. Let us now describe how this group acts on tensor decompositions.

Definition 2.2 *Let $\mathcal{T}$ denotes a rank-one tensor $\mathsf{A}\otimes\mathsf{B}\otimes\mathsf{C}$ and let g denotes the isotropy associated to matrices $(\mathsf{U}\times\mathsf{V}\times\mathsf{W})$ in $\mathrm{PSL}^{\pm}(\mathbb{R}^{m})\times\mathrm{PSL}^{\pm}(\mathbb{R}^{k})\times\mathrm{PSL}^{\pm}(\mathbb{R}^{n})$; the action $\mathsf{g}\diamond\mathcal{T}$ of g on $\mathcal{T}$ is the rank-one tensor:*

$$\big(\mathsf{U}^{-\intercal}\cdot\mathsf{A}\cdot\mathsf{V}^{\intercal}\big)\otimes\big(\mathsf{V}^{-\intercal}\cdot\mathsf{B}\cdot\mathsf{W}^{\intercal}\big)\otimes\big(\mathsf{W}^{-\intercal}\cdot\mathsf{C}\cdot\mathsf{U}^{\intercal}\big). \tag{15}$$

This action is extended by additivity to higher tensor rank tensors. Given two isotropies g_1 defined by matrices $(\mathsf{U}_1\times\mathsf{V}_1\times\mathsf{W}_1)$ and g_2 defined by matrices $(\mathsf{U}_2\times\mathsf{V}_2\times\mathsf{W}_2)$ in $\mathrm{PSL}^{\pm}(\mathbb{R}^{m})\times\mathrm{PSL}^{\pm}(\mathbb{R}^{k})\times\mathrm{PSL}^{\pm}(\mathbb{R}^{n})$, the composition $\mathsf{g}_1\circ\mathsf{g}_2$ is given by $(\mathsf{U}_1\cdot\mathsf{U}_2\times\mathsf{V}_1\cdot\mathsf{V}_2\times\mathsf{W}_1\cdot\mathsf{W}_2)$.

The following theorem recalls that all 2×2-matrix product algorithms with 7 coefficient multiplications are obtained by this single orbit of isotropies action on Strassen tensor decomposition:

Theorem 2.2 ([7, § 0.1]) *The group $\mathrm{PSL}^{\pm}(\mathbb{R}^{2\times2})^{\times3}$ acts transitively on the variety of fast algorithms multiplying 2×2-matrices.*

More generally, isotropy action on any decomposition may define another matrix product algorithm of same tensor rank but with potentially more interesting characteristics. This fact is used in [3, 5] to increase the numerical accuracy of some fast matrix multiplication algorithms. There, we use the tensor decomposition orbit to express $\langle 2\times2\times2{:}7\rangle$ and $\langle 3\times3\times6{:}40\rangle$ with more suitable coefficients. Let us first do the opposite in the following example.

Example 2.1 *Consider the following isotropy:*

$$\mathsf{g}=\left[\begin{smallmatrix}1&0\\0&I\end{smallmatrix}\right]\otimes\left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]\otimes\left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]. \tag{16}$$

The resulting trilinear representation associated to the tensor decomposition $\mathsf{g} \diamond \mathcal{S}$ *is:*

$$\begin{aligned}
&(a_{11} - I a_{22}) \otimes (b_{11} + b_{22}) \otimes (c_{11} + I c_{22}) && (17)\\
&+ (a_{12} + I a_{22}) \otimes (b_{21} + b_{22}) \otimes c_{11} && (18)\\
&+ (-a_{11} - I a_{21}) \otimes (b_{11} + b_{12}) \otimes c_{22} I && (19)\\
&+ (a_{11} + a_{12}) \otimes b_{22} \otimes (-c_{11} + c_{12}) && (20)\\
&+ a_{11} \, (b_{12} - b_{22}) \otimes (c_{12} + I c_{22}) && (21)\\
&- I a_{22} \, (-b_{11} + b_{21}) \otimes (c_{11} + I c_{21}) && (22)\\
&+ (-I a_{21} - I a_{22}) \otimes b_{11} \otimes (I c_{21} - I c_{22}) \,. && (23)
\end{aligned}$$

This last representation involves the complex number I but it is clear that there exists at least one point in the orbit of Strassen's tensor decomposition involving only integers.

So one can consider the following question:

Problem 2.1 *Is there a tensor decomposition in the orbit of* $\langle 3{\times}4{\times}7{:}63 \rangle$ *introduced in [10] that does not require complex numbers?*

We present now our main result in the following section.

3 A rational point in the orbit of AlphaEvolve's tensor decomposition

Using the Klein four stabilizer group (see [11]) of the original algorithm presented in [10], and educated guesses, we obtain the following isotropy:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \tag{24}$$

that sends the original complex-valued tensor decomposition to the following rational one:

$$m_1 = (a_{21} + a_{24}) \otimes (b_{17} - b_{37} + b_{47}) \otimes \left(\frac{c_{21}}{2} - \frac{c_{22}}{2} + \frac{c_{24}}{2} + \frac{c_{26}}{2} + \frac{c_{27}}{2} \right), \tag{25}$$

$$m_2 = \left(a_{11} - a_{21} + \frac{a_{12}}{2} - \frac{a_{22}}{2} \right) \otimes \left(\frac{b_{32}}{2} + \frac{b_{42}}{2} - \frac{b_{12}}{2} - b_{22} - b_{14} + b_{15} + b_{17} \right) \otimes (c_{16} + c_{17} - c_{12}), \tag{26}$$

$$m_3 = \left(\frac{a_{32}}{2} - \frac{a_{22}}{2} + a_{24} - a_{34} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} + b_{43} + b_{46} - b_{47} \right) \otimes (c_{31} + c_{34} + c_{37}), \tag{27}$$

$$m_4 = (a_{11} + a_{13}) \otimes \left(b_{32} + b_{15} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{12} + c_{15} - c_{17}), \tag{28}$$

$$m_5 = \left(\begin{array}{c} a_{14} + a_{34} \\ -\frac{a_{12}}{2} - \frac{a_{32}}{2} \end{array} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} + b_{43} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} + \frac{b_{44}}{2} + b_{46} - b_{47} \right) \otimes c_{34}, \tag{29}$$

$$m_6 = (a_{31} + a_{34}) \otimes \left(b_{11} + b_{46} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{31} + c_{36} + c_{37}), \tag{30}$$

$$m_7 = \left(a_{31} + \frac{a_{32}}{2} \right) \otimes (b_{16} - b_{36} - b_{46}) \otimes (-c_{16} + c_{26} + c_{36}), \tag{31}$$

$$m_8 = \left(a_{21} - a_{31} + \frac{a_{22}}{2} - \frac{a_{32}}{2} \right) \otimes \left(\frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17} \right) \otimes c_{26}, \tag{32}$$

$$m_9 = \left(\frac{a_{22}}{2} + \frac{a_{32}}{2} - a_{24} + a_{34} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} \right) \otimes (c_{21} + c_{31} + c_{34} + c_{37}), \tag{33}$$

$$m_{10} = \left(\frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{13} + a_{34} \right) \otimes \left(-b_{31} + b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{11} + c_{35}), \tag{34}$$

$$m_{11} = \left(\frac{a_{32}}{2} + a_{33} \right) \otimes (b_{13} - b_{33} + b_{43}) \otimes (c_{13} - c_{23} - c_{33}), \tag{35}$$

$$m_{12} =\left(\frac{a_{32}}{2} - \frac{a_{22}}{2} + a_{23} - a_{33}\right) \otimes \left(b_{31} + \frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} + b_{36} - b_{37}\right) \otimes (c_{33} - c_{35} + c_{37})\,, \tag{36}$$

$$m_{13} =\left(\frac{a_{32}}{2} + a_{34}\right) \otimes (b_{11} + b_{31} - b_{41}) \otimes (c_{11} - c_{21} - c_{31})\,, \tag{37}$$

$$m_{14} =\left(a_{21} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{13}\right) \otimes \left(b_{15} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (-c_{12} + c_{16} + c_{17} + c_{25})\,, \tag{38}$$

$$m_{15} =(a_{11} + a_{14}) \otimes \left(b_{42} - b_{14} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{12} - c_{14} - c_{17})\,, \tag{39}$$

$$m_{16} =\left(-a_{21} + a_{31} + \frac{a_{22}}{2} + \frac{a_{32}}{2}\right) \otimes \left(-\frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2}\right) \otimes (c_{26} - c_{32} + c_{36} + c_{37})\,, \tag{40}$$

$$m_{17} =\left(\begin{matrix} a_{11} + a_{31} \\ -\frac{a_{12}}{2} + \frac{a_{32}}{2} \end{matrix}\right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2}\right) \otimes (c_{32} - c_{16})\,, \tag{41}$$

$$m_{18} =\left(\begin{matrix} \frac{a_{12}}{2} + \frac{a_{32}}{2} \\ -a_{13} - a_{33} \end{matrix}\right)\otimes\left(b_{31} + \frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} - \frac{b_{15}}{2} + b_{25} - \frac{b_{35}}{2} - \frac{b_{45}}{2} + b_{36} - b_{37}\right) \otimes c_{35}, \tag{42}$$

$$m_{19} =\left(a_{11} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{23}\right) \otimes \left(b_{32} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{13} - c_{15} + c_{17} + c_{22})\,, \tag{43}$$

$$m_{20} =\left(a_{21} - a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2}\right) \otimes \left(b_{11} + b_{13} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17}\right) \otimes (c_{36} - c_{32} + c_{37})\,, \tag{44}$$

$$m_{21} =(a_{13} + a_{14}) \otimes \left(-b_{34} + b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (-c_{14} + c_{15} - c_{17})\,, \tag{45}$$

$$m_{22} =\left(\frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{33} - a_{14}\right) \otimes \left(b_{43} + b_{34} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (-c_{13} + c_{34})\,, \tag{46}$$

$$m_{23} =\left(-a_{11} + a_{21} + \frac{a_{12}}{2} - \frac{a_{22}}{2}\right) \otimes \left(\frac{b_{12}}{2} - b_{22} + \frac{b_{32}}{2} + \frac{b_{42}}{2} + b_{17}\right) \otimes c_{22}, \tag{47}$$

$$m_{24} =\left(-a_{11} + \frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{34}\right) \otimes \left(-b_{11} + b_{42} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{11} + c_{32})\,, \tag{48}$$

$$m_{25} =\left(a_{21} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{33}\right) \otimes \left(b_{13} - \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{23} + c_{32} - c_{36} - c_{37})\,, \tag{49}$$

$$m_{26} =\left(-\frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{13} + a_{24}\right) \otimes \left(b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{11} + c_{14} + c_{17} + c_{25})\,, \tag{50}$$

$$m_{27} =a_{32} \otimes\left(\frac{b_{41}}{2} - \frac{b_{13}}{2} - \frac{b_{11}}{2} - b_{21} - \frac{b_{31}}{2} - b_{23} + \frac{b_{33}}{2} - \frac{b_{43}}{2} + \frac{b_{16}}{2} - b_{26} - \frac{b_{36}}{2} - \frac{b_{46}}{2} + b_{27}\right)\otimes c_{37}, \tag{51}$$

$$m_{28} =\left(a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{23}\right) \otimes \left(b_{36} - \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{26} - c_{33} + c_{35} - c_{37})\,, \tag{52}$$

$$m_{29} =(a_{33} + a_{34}) \otimes \left(b_{31} + b_{43} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{31} + c_{33} + c_{37})\,, \tag{53}$$

$$m_{30} =\left(a_{21} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{14}\right) \otimes \left(-b_{14} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (-c_{12} + c_{16} + c_{17} - c_{24})\,, \tag{54}$$

$$m_{31} =\left(\begin{matrix} \frac{a_{12}}{2} + \frac{a_{32}}{2} \\ +a_{13} + a_{33} \end{matrix}\right)\otimes\left(\frac{b_{13}}{2} + b_{23} + \frac{b_{33}}{2} + \frac{b_{43}}{2} + b_{34} - \frac{b_{15}}{2} - b_{32} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} - b_{37}\right) \otimes c_{13}, \tag{55}$$

$$m_{32} =\left(-a_{11} + a_{21} + \frac{a_{12}}{2} + \frac{a_{22}}{2}\right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2}\right) \otimes (c_{12} - c_{16} - c_{17} + c_{22})\,, \tag{56}$$

$$m_{33} =\left(\frac{a_{22}}{2} - \frac{a_{32}}{2} + a_{23} - a_{33}\right) \otimes \left(\frac{b_{13}}{2} + b_{23} + \frac{b_{33}}{2} + \frac{b_{43}}{2} - b_{37}\right) \otimes c_{23}, \tag{57}$$

$$m_{34} =\left(\frac{a_{21}}{2} + \frac{a_{23}}{2}\right) \otimes (b_{17} + b_{37} - b_{47}) \otimes (-c_{22} + c_{23} - c_{25} + c_{26} + c_{27})\,, \tag{58}$$

$$m_{35} =\left(-a_{11} + \frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{33}\right) \otimes \left(b_{32} - b_{13} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{13} + c_{32})\,, \tag{59}$$

$$m_{36} =\left(\frac{a_{12}}{2} + \frac{a_{22}}{2} - a_{13} + a_{23}\right) \otimes \left(-\frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2}\right) \otimes (-c_{13} + c_{15} - c_{17} + c_{25})\,, \tag{60}$$

$$m_{37} =\left(\frac{a_{12}}{2} - \frac{a_{22}}{2} + a_{13} - a_{23}\right) \otimes \left(b_{32} - b_{34} + \frac{b_{15}}{2} - b_{25} - \frac{b_{35}}{2} + \frac{b_{45}}{2} + b_{37}\right) \otimes (c_{13} - c_{15} + c_{17}), \tag{61}$$

$$m_{38} =\left(\frac{a_{12}}{2} + \frac{a_{22}}{2} - a_{14} + a_{24}\right) \otimes \left(-\frac{b_{14}}{2} + b_{24} - \frac{b_{34}}{2} + \frac{b_{44}}{2}\right) \otimes (c_{11} + c_{14} + c_{17} + c_{24}), \tag{62}$$

$$m_{39} =\left(-\frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{33} + a_{24}\right) \otimes \left(-b_{43} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (-c_{23} + c_{31} + c_{34} + c_{37}), \tag{63}$$

$$m_{40} =a_{12} \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} - \frac{b_{44}}{2} - \frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} + b_{27}\right) \otimes c_{17}, \tag{64}$$

$$m_{41} =\left(-a_{11} + \frac{a_{12}}{2}\right) \otimes (-b_{12} + b_{32} + b_{42}) \otimes (c_{12} + c_{22} - c_{32}), \tag{65}$$

$$m_{42} =\left(\begin{array}{c} \frac{a_{12}}{2} + \frac{a_{32}}{2} \\ +a_{14} + a_{34} \end{array}\right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} + \frac{b_{41}}{2} - b_{42} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} - \frac{b_{44}}{2} - b_{45} - b_{47}\right) \otimes c_{11}, \tag{66}$$

$$m_{43} =\left(\frac{a_{12}}{2} - a_{13}\right) \otimes (b_{15} - b_{35} + b_{45}) \otimes (c_{15} + c_{25} - c_{35}), \tag{67}$$

$$m_{44} =\left(\frac{a_{12}}{2} - a_{14}\right) \otimes (b_{14} + b_{34} - b_{44}) \otimes (c_{14} + c_{24} - c_{34}), \tag{68}$$

$$m_{45} =\left(\begin{array}{c} -\frac{a_{12}}{2} + \frac{a_{32}}{2} \\ a_{13} + a_{33} \end{array}\right) \otimes \left(\frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} - \frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2}\right) \otimes (-c_{13} + c_{35}), \tag{69}$$

$$m_{46} =\left(a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{24}\right) \otimes \left(-b_{46} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (-c_{26} + c_{31} + c_{34} + c_{37}), \tag{70}$$

$$m_{47} =\left(\frac{a_{22}}{2} + \frac{a_{32}}{2} - a_{23} + a_{33}\right) \otimes \left(\frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2}\right) \otimes (c_{23} + c_{33} - c_{35} + c_{37}), \tag{71}$$

$$m_{48} =\left(\begin{array}{c} -a_{11} - a_{31} \\ \frac{a_{12}}{2} + \frac{a_{32}}{2} \end{array}\right) \otimes \left(b_{11} - \frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + b_{13} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17}\right) \otimes c_{32}, \tag{72}$$

$$m_{49} =\left(-\frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{23} + a_{14}\right) \otimes \left(-b_{34} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{13} - c_{15} + c_{17} - c_{24}), \tag{73}$$

$$m_{50} =\left(a_{31} + \frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{14}\right) \otimes \left(b_{14} + b_{46} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (-c_{16} + c_{34}), \tag{74}$$

$$m_{51} =\left(-\frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{23} + a_{34}\right) \otimes \left(b_{31} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{21} - c_{33} + c_{35} - c_{37}), \tag{75}$$

$$m_{52} =\left(\frac{a_{22}}{2} - \frac{a_{32}}{2} + a_{24} - a_{34}\right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} + \frac{b_{41}}{2} - b_{47}\right) \otimes c_{21}, \tag{76}$$

$$m_{53} =\left(\frac{a_{12}}{2} - \frac{a_{22}}{2} - a_{13} + a_{23}\right) \otimes \left(\frac{b_{15}}{2} - b_{25} + \frac{b_{35}}{2} + \frac{b_{45}}{2} + b_{37}\right) \otimes c_{25}, \tag{77}$$

$$m_{54} =\left(\begin{array}{c} \frac{a_{32}}{2} - \frac{a_{12}}{2} \\ +a_{14} + a_{34} \end{array}\right) \otimes \left(-\frac{b_{11}}{2} - b_{21} - \frac{b_{31}}{2} + \frac{b_{41}}{2} - \frac{b_{14}}{2} + b_{24} - \frac{b_{34}}{2} + \frac{b_{44}}{2}\right) \otimes (c_{11} + c_{34}), \tag{78}$$

$$m_{55} =\left(a_{11} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{24}\right) \otimes \left(b_{42} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2}\right) \otimes (c_{11} + c_{14} + c_{17} + c_{22}), \tag{79}$$

$$m_{56} =\left(\frac{a_{12}}{2} - \frac{a_{22}}{2} - a_{14} + a_{24}\right) \otimes \left(\frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} + \frac{b_{44}}{2} - b_{47}\right) \otimes c_{24}, \tag{80}$$

$$m_{57} =\left(\frac{a_{23}}{2} + \frac{a_{24}}{2}\right) \otimes (-b_{17} + b_{37} + b_{47}) \otimes (c_{21} + c_{23} + c_{24} - c_{25} + c_{27}), \tag{81}$$

$$m_{58} =\left(a_{21} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{34}\right) \otimes \left(b_{11} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{21} + c_{32} - c_{36} - c_{37}), \tag{82}$$

$$m_{59} =\left(\begin{array}{c} a_{11} + a_{31} \\ +\frac{a_{12}}{2} + \frac{a_{32}}{2} \end{array}\right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + b_{14} - b_{15} + \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17}\right) \otimes c_{16}, \tag{83}$$

$$m_{60} =\left(a_{31} + \frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{13}\right) \otimes \left(b_{15} - b_{36} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2}\right) \otimes (c_{16} + c_{35}), \tag{84}$$

$$m_{61} =(a_{31}+a_{33}) \otimes \left(b_{13}+b_{36}-\frac{b_{17}}{2}-\frac{b_{37}}{2}+\frac{b_{47}}{2}\right) \otimes (c_{33}+c_{36}+c_{37})\,, \tag{85}$$

$$m_{62} =\left(\frac{a_{12}}{2}-\frac{a_{22}}{2}+a_{14}-a_{24}\right) \otimes \left(b_{42}-\frac{b_{14}}{2}+b_{24}-\frac{b_{34}}{2}+\frac{b_{44}}{2}+b_{45}+b_{47}\right) \otimes (c_{11}+c_{14}+c_{17})\,, \tag{86}$$

$$m_{63} =a_{22} \otimes b_{27} \otimes c_{27}. \tag{87}$$

The type of this tensor decomposition is:

$$6X^2Y^3Z^2 + 15X^2Y^2Z^2 + 2XY^4Z + 6X^2YZ^2 + 12XY^3Z + 12XY^2Z + 10XYZ. \tag{88}$$

Several other representations of this algorithm are given in the following section, in appendix and are available in [11, 2].

4 Straight-line programs for the rational $\langle 3\times 4\times 7{:}63\rangle$

We here give some (PLinOpt generated [2]) straight-line programs obtained from L, R, P matrices presented in Appendix B and associated to the tensor decomposition given by Equations (25) to (87):

Listing 1: SLP for input linear forms defined by matrix L

```
r12:=A12/2; l40:=r12-A11; r14:=A32/2; l0:=A21+A24; l6:=A31+r14; l10:=A33+r14;
l22:=A21-A22/2+l40; l26:=A32; l28:=A33+A34; l33:=(A21+A23)/2; l42:=r12-A13;
l43:=r12-A14; t63:=l40-l22; t64:=A22-t63; t65:=l28-l10; l12:=t65+l26; u67:=l33*2;
t67:=u67-t64; t68:=l0-t64; t69:=A12-l42; t70:=t63+u67; t71:=t63+l0; t72:=l10-l26;
l16:=l6-l40; t74:=A12-l43; l31:=l22+A22; l15:=l6+t63; l44:=l10-l42; l27:=l6+t67;
l2:=t68-t65; l9:=l42+l12; l13:=t64-l42; l53:=l12-l43; l47:=l26-l16; l21:=l10+l43;
l17:=l42-t72; l11:=t67-t72; l51:=t71-l12; l34:=l10+l40; l25:=t71-l42; l52:=l42+t67;
l14:=t74-l40; l37:=l43+t71; l36:=t69-t70; l57:=l12-t63; l35:=l42+t70; l18:=u67-l22;
l1:=A12-l31; l50:=t67+l12; l23:=l40+l12; l38:=l10+t68; l41:=l12+t74; l61:=t74-t71;
l4:=t65-l43; l24:=l10-t63; l59:=l6+l42; l7:=t64-l6; l45:=l6+t68; l55:=l43+t68;
l54:=l0-l22; l48:=t70-l43; l49:=l6+l43; l32:=t70-l10; l8:=l12-t68; l5:=l6+t65;
l46:=l10-t67; l60:=l6+t72; l58:=A12+l16; l30:=l10+t69; l19:=l26-l15; l29:=t64-l43;
l3:=t69-l40; l20:=t69-l43; l56:=l33+t63+(l0-A22)/2; l39:=A12; l62:=A22;
```

Listing 2: SLP for input linear forms defined by matrix R

```
y28:=B17-B37; r56:=B47-y28; r0:=B47+y28; r33:=B17+B37-B47; y33:=B14+B34; y34:=B13+B43;
y35:=B11+B31; y36:=B32+B42; y37:=B36+B46; y38:=B15+B45; y39:=B24-y33/2; y40:=B21+y35/2;
y41:=B25-y38/2; y42:=B22-y36/2; y43:=B26+y37/2; y44:=B23+y34/2; b45:=B33/2;
r46:=y44-b45; b46:=B35/2; r35:=y41+b46; b47:=B41/2; r8:=y40-b47; b48:=B12/2;
r31:=y42+b48; b49:=B44/2; r37:=y39+b49; b50:=B16/2; r15:=y43-b50; b51:=r56/2;
y51:=b51-B34; b52:=r0/2; r45:=b52-B46; r50:=B31-b51; r54:=B42+b52; b55:=r33/2;
r18:=B32+b55; r27:=B36-b55; r32:=y44+b45-B37; y64:=B42+B45+r37; y68:=y41-b46;
r51:=b47+y40-B47; y72:=r35+B34-B32; y74:=y42-b48; y75:=B46+B43+r8; r7:=b50+y43-B17;
r19:=B11+B13+r15-B17; y78:=B15-B14-r31; r55:=b49-B47-y39; r11:=r46+B36+B31-B37;
r1:=B17+y78; r2:=y75-B47; r3:=B15+r18; r4:=y75+r55; r5:=B11-r45; r6:=B16-y37;
r9:=B45-r50; r10:=y34-B33; r12:=y35-B41; r13:=B15+b55; r14:=r54-B14; r16:=r31+r15;
r17:=y68+r11; r20:=B45+y51; r21:=B43-y51; r22:=B17-y74; r23:=r54-B11; r24:=B13-b55;
r25:=B45+b51; r26:=B27-r46-r8-r15; r28:=B43+r50; r29:=b52-B14; r30:=r32+y72;
r34:=r18-B13; r36:=B37-y72; r38:=b51-B43; r39:=B27+r35+r31-r37; r40:=y36-B12;
r41:=r51-y64; r42:=y38-B35; r43:=y33-B44; r44:=r46+r35; r47:=y74+r19; r48:=y51;
r49:=B14-r45; r52:=B37-y68; r53:=r37-r8; r57:=B11-b52; r58:=r7-y78; r59:=B15-r27;
r60:=B13+r27; r61:=B47+y64; r62:=B27;
```

Listing 3: products

```
p0:=l0*r0; p1:=l1*r1; p2:=l2*r2; p3:=l3*r3; p4:=l4*r4; p5:=l5*r5; p6:=l6*r6; p7:=l7*r7;
p8:=l8*r8; p9:=l9*r9; p10:=l10*r10; p11:=l11*r11; p12:=l12*r12; p13:=l13*r13;
p14:=l14*r14; p15:=l15*r15; p16:=l16*r16; p17:=l17*r17; p18:=l18*r18; p19:=l19*r19;
p20:=l20*r20; p21:=l21*r21; p22:=l22*r22; p23:=l23*r23; p24:=l24*r24; p25:=l25*r25;
p26:=l26*r26; p27:=l27*r27; p28:=l28*r28; p29:=l29*r29; p30:=l30*r30; p31:=l31*r31;
p32:=l32*r32; p33:=l33*r33; p34:=l34*r34; p35:=l35*r35; p36:=l36*r36; p37:=l37*r37;
p38:=l38*r38; p39:=l39*r39; p40:=l40*r40; p41:=l41*r41; p42:=l42*r42; p43:=l43*r43;
p44:=l44*r44; p45:=l45*r45; p46:=l46*r46; p47:=l47*r47; p48:=l48*r48; p49:=l49*r49;
p50:=l50*r50; p51:=l51*r51; p52:=l52*r52; p53:=l53*r53; p54:=l54*r54; p55:=l55*r55;
p56:=l56*r56; p57:=l57*r57; p58:=l58*r58; p59:=l59*r59; p60:=l60*r60; p61:=l61*r61;
p62:=l62*r62;
```

Listing 4: SLP for input linear forms defined by matrix P

```
q12:=p56+p33; q20:=p61+p54+p37+p25; q29:=p48+p36-p35+p18; q34:=p57+p24-p19-p15;
q40:=p50-p46+p27-p11; q53:=p45+p38+p8+p2; q54:=p29-p31+p13+p1; z37:=p0/2; z36:=p56+z37;
q55:=p33+z37; C24:=p55-p48+p43+p37-p29+z36; C21:=p57+p51+p50-p12+p8+z36;
C25:=p52+p42+p35+p25+p13-q12; C26:=p27-p45+p15+p7+p6+q55; C22:=p54+p40+p31+p22+p18-q55;
C27:=p62+z37+q12; C23:=p46-p38+p32+p24-p10+q12; z23:=p5+q53;
C34:=p53+p49-p43+p21+p4+q53; q56:=q20-p20-p14; C11:=p53+p41+p23+p12+p9+q20;
q57:=p60+p28-q40; C35:=p59+p44-p42+p17+p9+q40; C32:=p47-p40+p34+p23+p16+q34;
q58:=q54-p3; C16:=p59+p58-p49-p16-p6+q54; C36:=p60+p6+p5-q34; C15:=p42+p20+p3-q29;
C13:=p34-p44+p30-p21+p10+q29; C12:=p40+p14-q58; C31:=p28-p12+z23; C14:=p43+q56;
C17:=p39+q29+q58+q56; C33:=q57-p10; C37:=p26-q34+z23+q57;
```

These straight-line programs require:

- 73 additions and 6 multiplications (binary shifts) for `L`;
- 98 additions and 15 multiplications (binary shifts) for `R`;
- 107 additions and 1 multiplication (binary shifts) for `P`.

5 Concluding remarks

This note concludes successfully a work started in [4] whose purpose is to simplify the complex-valued fast matrix multiplication algorithms introduced in [10].

As done in [4] for the rational $\langle 4{\times}4{\times}4{:}48\rangle$ tensor decomposition, we could explicit the leading constant of asymptotic complexity of the $\langle 3{\times}4{\times}7{:}63\rangle$ tensor decomposition presented in this note. This would use the number of operations of the associated SLPs, given in Section 4. But, to do so we have to consider square matrices, and the natural candidate to such a study should be the tensor decomposition $\langle 84{\times}84{\times}84{:}250057\rangle$.

Now, as 84 is equal to $3 \cdot 4 \cdot 7$ and 250057 is equal to 63^3, such a tensor decomposition is obtained by the tensor product of $\langle 3{\times}4{\times}7{:}63\rangle$ with $\langle 7{\times}3{\times}4{:}63\rangle$ and $\langle 4{\times}7{\times}3{:}63\rangle$. Up to Tellegen's transposition principle, the other two tensor decompositions would share the same L, R, P matrices (permuted and transposed) and thus similar SLPs. This would then give an algorithm with cost $O\big(n^{3\log_{84} 63}\big)$, that is $O\big(n^{2.805217355}\big)$.

Nevertheless, as shown in [1], there exists a tensor decomposition $\langle 84{\times}84{\times}84{:}226100\rangle$ with a better asymptotic cost: $O\big(n^{\log_{84} 226100}\big)$ that is $O\big(n^{2.782496596}\big)$.

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[13] Volker Strassen. Gaussian elimination is not optimal. *Numerische Mathematik*, 13(4):354–356, August 1969. doi:10.1007/BF02165411. (cited in sec. 1)

A Source codes

All the different matrices presented in this appendix can be found in the PLinOpt library's data directory [2]. For straight line programs presented in Appendix B:

- `data/3x4x7_63_rational_L.sms`;
- `data/3x4x7_63_rational_R.sms`;
- `data/3x4x7_63_rational_P.sms`.

For alternative basis based straight line programs presented in Appendix C:

- `data/3x4x7_63_rational-ALT_L.sms`, `data/3x4x7_63_rational-CoB_L.sms`;
- `data/3x4x7_63_rational-ALT_R.sms`, `data/3x4x7_63_rational-CoB_R.sms`;
- `data/3x4x7_63_rational-ALT_P.sms`, `data/3x4x7_63_rational-CoB_P.sms`.

B LRP representation of the algorithm

Notation B.1 *Given an* $m\times k$*-matrix* A*, we denote by* $\mathsf{A}_{i,*}$ *the ith row and by* $\mathrm{Vect}(\mathsf{A})$ *the row-major vectorization of this matrix, i.e. the vector* v *in* $\mathbb{R}^{mk}$ *of the matrix coefficients such that* $v_{ik+j} = a_{i,j}$*.*

Definition B.1 *The* LRP *representation of a tensor decomposition* $T = \sum_{i=1}^{r} \mathsf{M}_i \otimes \mathsf{N}_i \otimes \mathsf{O}_i$ *in the tensor product space* $\left(\mathbb{R}^{m\times k}\right)^{\star} \otimes \left(\mathbb{R}^{k\times n}\right)^{\star} \otimes \mathbb{R}^{m\times n}$*, is formed by the three matrices* L *in* $\mathbb{R}^{r\times mk}$*,* R *in* $\mathbb{R}^{r\times kn}$*, and* P *in* $\mathbb{R}^{mn\times r}$*, such that* $\mathsf{L}_{i,*}$ *is* $\mathrm{Vect}(\mathsf{M}_i)$*,* $\mathsf{R}_{i,*}$ *is* $\mathrm{Vect}(\mathsf{N}_i)$ *and* $\mathsf{P}_{*,i}$ *is* $\mathrm{Vect}(\mathsf{O}_i{}^{\intercal})$*.*

The isotropies action on a LRP representation is a direct consequence of results presented in Section 2.2 and is summarized in the following lemma.

Lemma B.1 *Let* g *be three matrices* $(\mathsf{U}\times\mathsf{V}\times\mathsf{W})$ *in* $\mathrm{PSL}^{\pm}(\mathbb{R}^m)\times\mathrm{PSL}^{\pm}(\mathbb{R}^k)\times\mathrm{PSL}^{\pm}(\mathbb{R}^n)$ *and a* LRP *representation* $[\mathsf{L};\mathsf{R};\mathsf{P}]$ *of a matrix product tensor decomposition, the action* $\mathsf{g}\diamond[\mathsf{L};\mathsf{R};\mathsf{P}]$ *of* g *on* $[\mathsf{L};\mathsf{R};\mathsf{P}]$ *is another* LRP *representation of a matrix product tensor decomposition defined by:*

$$\left[\mathsf{L}\cdot\left(\mathsf{V}^{\intercal}\otimes\mathsf{U}^{-1}\right);\mathsf{R}\cdot\left(\mathsf{W}^{\intercal}\otimes\mathsf{V}^{-1}\right);\left(\mathsf{U}\otimes\mathsf{W}^{-\intercal}\right)\cdot\mathsf{P}\right]. \tag{89}$$

We present now the LRP representation of the tensor decomposition given by Equations (25) to (87):

$$L=\begin{bmatrix}
0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
1 & \frac{1}{2} & 0 & 0 & -1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 0 & -1\\
1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 & -1 & -\frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & 1\\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & -1 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
0 & -\frac{1}{2} & 1 & 0 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -1 & 0\\
1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & -1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1\\
0 & -\frac{1}{2} & 0 & 1 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0\\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
0 & \frac{1}{2} & -1 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 1 & 0\\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 & 0 & \frac{1}{2} & 1 & 0\\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0\\
0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & 0 & 1\\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & -\frac{1}{2} & 0 & -1\\
0 & \frac{1}{2} & -1 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0\\
0 & \frac{1}{2} & 0 & -1 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 1\\
1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0\\
0 & \frac{1}{2} & 0 & 1 & 0 & -\frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix},R=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1\\
0 & -\frac{1}{2} & 0 & -1 & 1 & 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0\\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 1 & -1\\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
\frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & \frac{1}{2} & 0 & 1 & -1\\
1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0\\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2}\\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & 0 & 1 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0\\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0\\
0 & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0\\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & 1 & -1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
1 & 0 & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 & 0 & -\frac{1}{2}\\
0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0\\
-1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 1 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2}\\
-\frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & -1 & 0 & -1 & 0 & 0 & -1 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 & 0 & -\frac{1}{2}\\
0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & \frac{1}{2} & 1 & \frac{1}{2} & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0\\
0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1\\
0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & -1 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0\\
0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2}\\
0 & \frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 1 & 0 & -1 & 1 & 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0\\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\
\frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 & -\frac{1}{2} & -1 & 0 & -1\\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0\\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & -1 & \frac{1}{2}\\
0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0\\
1 & -\frac{1}{2} & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & -1\\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0\\
-\frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -1\\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1\\
1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
0 & \frac{1}{2} & 0 & 1 & -1 & \frac{1}{2} & -1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0\\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2}\\
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0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 1 & 0 & 1\\
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C Alternative bases algorithm

Following [8], we present in this section the L, R, P matrices of an alternative basis derived from the tensor decomposition given by Equations (25) to (87):

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\end{bmatrix}, \tag{91}$$

$$P_{\text{alt}} = \begin{bmatrix}
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0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0
\end{bmatrix}. \tag{92}$$

The (PLinOpt generated) SLPs associated to these L, R, P matrices can be realized with straight-line programs with, respectively, $1, 1$ and 2 additions.

Then their respective change of basis matrices are given thereafter, whose associated straight-line programs require less than:

- 72 additions and 5 multiplications (binary shifts) for `CoB_L`;
- 97 additions and 15 multiplications (binary shifts) for `CoB_R`;
- 105 additions and 1 multiplication (binary shift) for `CoB_P`.

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & -1 & 0 \\
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{1}{2} & 0 & 1 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
0 & \frac{1}{2} & -1 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 \\
0 & -\frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
0 & \frac{1}{2} & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 & -1 & -\frac{1}{2} & 0 & 0 \\
0 & \frac{1}{2} & 0 & -1 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 1 & \frac{1}{2} & 0 & 0 \\
0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 \\
1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
0 & \frac{1}{2} & 0 & 1 & 0 & -\frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & -\frac{1}{2} & 1 & 0 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & \frac{1}{2} & -1 & 0 & 0 & \frac{1}{2} & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 1 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 \\
-1 & \frac{1}{2} & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & \frac{1}{2} & 0 & 0 & -1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & -\frac{1}{2} & 0 & -1 \\
0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 \\
1 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & \frac{1}{2} & 0 & -1 \\
0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 & 1 \\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 1 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
-1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 1 \\
1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{2} & 0 & -1 & 0 & 0 & 0 & 0 & 1 & \frac{1}{2} & 0 & 0 \\
-1 & \frac{1}{2} & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\quad
\begin{bmatrix}
0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 1 & -1 \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & \frac{1}{2} & 1 & \frac{1}{2} & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 1 & 0 & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 \\
0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & 0 & 1 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\
\frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & -1 & 0 & -\frac{1}{2} & -1 & 0 & -1 \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\
0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & 1 & -1 & 0 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 \\
-\frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & -1 & 0 & -1 & 0 & 0 & -1 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 \\
0 & -\frac{1}{2} & 0 & -1 & 1 & 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\
0 & \frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 1 & 0 & -1 & 1 & 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & \frac{1}{2} & 0 & 1 & -1 & \frac{1}{2} & -1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
1 & 0 & 1 & 0 & 0 & -\frac{1}{2} & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\
0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -1 \\
0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 1 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} \\
0 & 0 & 0 & 0 & 1 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\
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